

ATRC: An Anonymous Traceable and Revocable Credential System Using Blockchain for VANETs

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Abstract—With the number of smart vehicle drivers increasing rapidly, privacy-preserving identity management methods in Vehicular Ad-hoc Networks (VANETs) become more imperative and receive much attention from researchers. Since identity leakage or a single point of failure may result in serious consequences in the VANET, the decentralized anonymous credential (DAC) could be a potential approach to construct a robust network. However, the supervision under decentralized environment could be a necessary and troublesome in the VANETs. In this paper, we present an anonymous traceable and revocable credential system using blockchain, called ATRC, built over a generalized group signature. The underlying group signature not only meets the anonymity requirement but also makes the users master their identities. What's more, to tackle the privacy leakage in the revocation process, we employ the Merkle tree to construct a whitelist, which trades off the efficiency and the privacy. Finally, we give the comparison and experiment performance to show our scheme holds lightweight on the user side and has fewer computation costs in the show and revocation phase.

Index Terms—VANETs, decentralized anonymous credential, group signature, blockchain.

I. INTRODUCTION

NOWADAYS drive becomes one of the main transportation ways in cities, which prompts the research of smart vehicles. Smart vehicles, which are equipped with some computation devices, are active roles in the Intelligent Transport Systems (ITS). For an information system with a large user group, the seriousness of identity privacy leakage could be much great. As a representative incident, 77 million account information of PlayStation network subscribers were disclosed, which was caused by the external intrusion [1], [2]. However, privacy leakage could be more serious in the VANET environment, since there are many legal disputes or even fatal road accidents. Compared to traditional central systems, the credential-focus system means that more data are controlled by vehicle users in order to protect sensitive information, which makes it much flexible for vehicle users to take charge of their assets via customized strategy [3]. What's more, storing data on the vehicle user side could avoid most storage costs of certificate issuance authority, therefore the ITS could be more robust and the seriousness of identity leakage may be relieved.

In the VANET environment, the communication entities including the On-Board Unit (OBU) and the Road-Side Unit (RSU) could afford complex operations. Also, the communications mainly include Vehicular-to-Infrastructure (V2I) and Vehicle-to-Vehicle (V2V). The smart vehicles equipped with OBU could get the road information with the infrastructure like RSU after the identity authentication process. After the records of bad behavior are reported to the revocation authority, the certificates of misbehaving vehicles will be revoked. Although the revocation by revealing identities may get some convenience, it practically damages the identity privacy of revoked vehicle users. The architecture of the VANET is shown in Fig. 1.

However, the revocation operation in credential systems for VANETs may comprise the privacy of the vehicle user as well as the efficiency of the communication. In detail, two credential revocation approaches are practical in many related schemes [4]. The first is verifier-local revocation, which usually employs a blacklist or whitelist as the restriction [5]. Especially, both of them need a trusted third party (TTP) and the vehicle users have no extra update operation in the revocation except the membership proof in verification. The second is accumulator revocation that witnesses are generated and distributed for all vehicle users [6]. Then all vehicle users are supposed to update their witness after the accumulator updates. For VANETs, the window time between the report of misbehaving vehicle and the publication of the new revoke list should be as little as possible,

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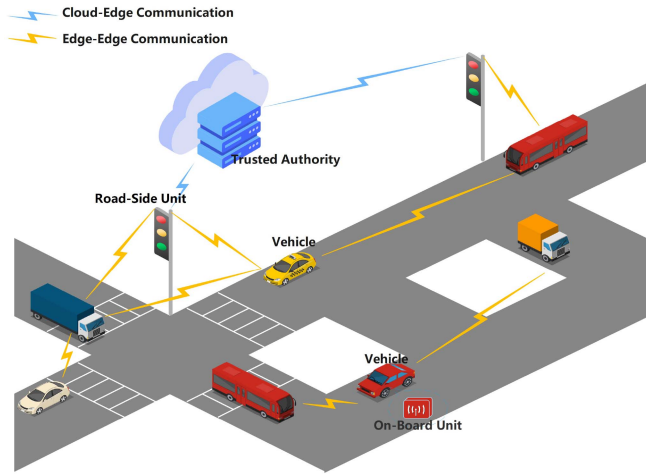


Fig. 1. Architecture of VANET.

since the misbehaving vehicle might cause a new accident before revocation. Thus it is reasonable to take the window time of revocation into consideration.

The group signature is a special kind of the digital signature [7], [8], which allows members to generate signatures on behalf of the group, is one of the key components of the cryptographic primitive. On the one hand, group signatures could hide the identity of a specific signer [9], [10], which accomplishes anonymousness in practice. On the other hand, the signer could not escape from the responsibility of malicious behavior since group signature makes it convenient for the group manager to trace a specific member in theory. In fact, it is very practicable to employ the group signature in credential systems for anonymousness and traceability.

Inspired by the techniques of Ateniese et al. [11], the vehicle user regenerates his own pseudonyms locally after the register phase, while the TTP management authority only takes part in the register and revocation phase. In our scheme, the management authority takes the duty of group manager. The pseudonym, which is actually a group signature generated by a vehicle user, not only hides the user's identity information but also is a clue in the trace phase. In practice, the vehicle user could change his pseudonym to hide trail for others except for the management authority. Based on the framework proposed by Garman et al. [12], we extend the basic process with the revocation phase, which is mainly carried out by the management authority. To address the huge amount of computation in the whitelist, we employ a Merkle tree to reduce the complexity of computation on the management authority side. With the Merkle tree, the management authority is supposed to update the path of revoked user and uploads the new root to the blockchain for public access while the cloud server can take charge of path query operation.

A. Contributions

Our main contributions are presented as follows.

- 1) In this paper, we present a high-level description of anonymous traceable and revocable credential systems for VANET, which shows the approach to constructing

these systems using generalized group signatures as well as customized revocation methods.

- 2) Moreover, we propose a new privacy-preserving revocation method using the Merkle tree. In the revocation phase, misbehaving vehicle users are revoked quickly and still keep anonymous. In addition, the burden of updating the whitelist is affordable to the management authority.
- 3) Finally, inspired by work in Ateniese et al. [11], we provide an instance scheme and implement the system, which holds feasible computation costs in the show and revocation phase.

B. Organization

We organize this paper as follows. In Section II, we will survey and analyze the relate works of anonymous credential system. In Section III, some necessary preliminaries will be introduced. Our proposed system will be presented and illustrated in Section IV and the security analysis is presented in Section V. Correspondingly, in Section VI performance evaluation will be displayed. Finally, conclusion is drawn in Section VII.

II. RELATED WORK

Anonymous credential, which is first proposed by Chaum [13], vastly improves the researches of privacy-preserving identity management systems. Camenisch and Lysanskaya [14] proposed a signature scheme that could be referred to construct anonymous credential systems. Specially, this scheme employs bilinear pairing, which will make it flexible for further improvement. As a practical and flexible approach for privacy-preserving, the anonymous credential is adapted in many fields such as IoT [15] and the cloud [16]. Tan and Groß [17] proposed a new attribute-based anonymous credential, which is especially efficient in unrestricted attribute space. This scheme modifies the arrangement of attributes in credentials to implement an access policy. Attribute accumulator is also valuable for reference. Nakanishi and Kanatani [18] proposed a blacklist table anonymous credential with a pairing-based accumulator, which authentication progress holds the constant size of authentication data without a trusted third party. While central authority is common in traditional identity management systems, trusted problems about privacy and single point of failure are the main potential risks. What's more, after the bitcoin [19] becomes popular, distributed systems like blockchain are employed in many research works to make systems decentralized and robust [20], [21].

The framework of decentralized anonymous credential is proposed by Garman et al. [12] for removing the existent of the trusted credential issuer in anonymous credential systems. Then plenty of schemes inspired by this framework have been proposed. Sonnino et al. [22] proposed a special selective disclosure credential solution based on the blind signature and bilinear pairings, which holds a short and very efficient credential. Halpin [23] was inspired by Sonnino's work [22], then he improved this work by transforming the original token into a well-designed token to append scalability of embedding arbitrary attributes. Lin et al. [24] proposed a privacy-enhancing

decentralized anonymous credentials system with efficient range proofs aimed at compromising the full user attributes list in the credential show phase. Cui et al. [25] proposed an anonymous and traceable data sharing protocols based consortium blockchain, which makes the combination of the consortium blockchain and the base station to avoid the RSU. However, the situation could be limited as the vehicles hold high movability. Yang et al. [26] presented a trust model about the vehicle networks, which constructs the trust among the vehicles and the punishment could damage the trust. What's more, the supervision could be weak. Inedjaren et al. [27] combined the VANET and blockchain by applying the optimized link state routing protocol to solve the potential risks. However, the efficiency could be unpractical since the block is collaboratively generated by the nodes. Das et al. [28] proposed a secure blockchain-enabled V2V communication system, called BVCS, which is aimed to improve the data sharing between vehicles. What's more, the lack of security model could be the block of the practical usage.

However above skillful solutions don't contain revocation, which is an important feature in credentials systems. With the revocation function, malicious behavior could be kept within limits.

Since the revocation function could benefit long-time running, many solutions have been proposed in different revocation ways. Derler et al. [5] designed a new revocable multishow attribute-based anonymous credential building on previous work, which holds well scalability and the constant work of two roles. In this system, the accumulator is adapted to construct a blacklist for the revocation. Yang et al. [4] proposed a lightweight anonymous authentication based on bilinear pairings with the function of outsourced witness update aimed at the bottleneck of anonymous credentials. The revoked key will be revealed in the revocation phase implemented by accumulators. Bui and Aura [29] constructed a framework of the revocation in distributed access control to conveniently operate the revocation methods. In detail, during the revocation, the public keys of revoked users are revealed. Ma et al. [30] proposed an attribute-based encryption algorithm using blockchain, which could provide the preserving-privacy protocols as well as the audit methods. However, the revocation is not supported.

As mentioned above, the revocation with accumulator could either increase the update burden of unrevoked users or reveal revoked user's identities. In this case, it's not reasonable to make unrevoked users take the burden of revocation and permanently get rid of some identity for lapses. Maybe, it is a better choice to keep revoked identity private and disabled.

III. PRELIMINARIES

A. Decentralized Anonymous Credential

In general, the DAC system runs around credentials. For the credential issuance, an organization will take charge of issuance policies since there is no single issuance party in the DAC systems [12], [31], [32]. In detail, an organization consists of a group of users in this decentralized system. The term "organization" remains consistent with previous work and is a little different in meaning.

In terms of the DAC framework, we will give a brief introduction as follows.

- *Setup*: To set up this system, this algorithm receives the security parameter λ as input, then it returns the public parameters pp as output.
- *KeyGen*: To generate the key for users, this algorithm receives the public parameter pp as input, then it returns user's secret key sk_U as output.
- *PIDForm*: In order to form the pseudonyms for users, this algorithm receives the public parameter pp and user's secret key sk_U as input, then it returns user's pseudonyms PID according to input key sk_U as output.
- *CredForm*: In order to form the credentials related to pseudonyms, this algorithm receives user's secret key sk_U , the user's pseudonyms PID , the attributes $attrs$ and the auxiliary aux as input, then it returns the credential c and a proof π_M as output. The proof π_M shows the relationship between the pseudonyms PID and the credential c . The attributes $attrs$ is included in credential c . Then the results will be uploaded to the blockchain.
- *CredVerf*: To verify the credential, this algorithm receives user's pseudonyms PID , attributes $attrs$, auxiliary aux , credential c and proof π_M as input, then it returns 1 if π_M holds.
- *Show*: In order to show the qualification of credential, this algorithm receives the secret key sk_U , the pseudonyms PID and the credential c as input, then it returns a proof π_S as output. The proof π_S shows the qualification of the credential.
- *ShowVerf*: To verify the shown data, this algorithm receives proof π_S and credential c as input, then it returns 1 if π_S holds.

The decentralized anonymous credential is flexible. Credential verification could be performed publicly as the policy of credentials is accessible and consistent. Moreover, the attributes in credentials could be organized by the user to achieve minimal necessary reveal for privacy-preserving.

B. Group Signature

The group signature is first proposed in 1991. The main idea is that members in the group could generate a group signature, others outside this group could verify this signature without knowing the actual signer. In detail, the group is managed by the group manager, who will take charge of the registration of group members. Especially, group manager holds the ability to trace malicious group members. A brief introduction will be given as follow.

- *Setup*: The group manager sets up a group, a group key *secretkey* and other parameters *params*.
- *Join*: The group manager communicates with the user who wants to join this group. The user will become a group member with the group key and the group certificate *cert* if the group manager admits.
- *Sign*: The group member could sign messages representing this group by using the group key and certificate *cert*. Then it outputs the signature *sig* corresponding to the message.

- *Verf*: Others outside the group could verify group signature with group public key with knowing nothing except validation of signature *sig*.
- *Open*: For a valid group signature, the group manager could trace the signer by using the group key *secretkey*.

The group signature has been adapted in many fields for the privacy-preserving and traceability. Actually, it may be more practical if solutions employ group signature with revocation, which attracts many researchers' attention and plenty of solutions have been proposed.

C. Merkle Tree

The Merkle tree is improved as a complete binary tree. Nowadays, the Merkle tree is still an important and practical technology in authentication and integrity checking [33], [34]. What's more, researchers paid much attention to the Merkle tree [35], [36]. Especially, the Merkle tree is employed in blockchain to make block lightweight and convenient for verification.

In blockchain systems, the Merkle tree is very common and efficient [37]. Merkle tree provides a useful approach to quickly check the completeness of the transactions in the block since the root hash contains all information of the transactions. The special tree struct makes it possible that locate the wrong message in the leaf node without scanning every leaf nodes sequentially. What's more, the Merkle tree could play a role in the member proof. While leaf nodes represent members, the root hash could reflect the whole members. In addition, hash string hides the real value of user identity and the smart contracts [38] could provide powerful functions, which may be valuable to construct a privacy-enhance solution.

IV. OUR PROPOSED SYSTEM

A. System Model

The proposed system is composed of vehicle users, the management authority(MA), the RSU, and the blockchain. The whole system is shown in Fig. 2. The four parties are displayed below:

- *Vehicle Users*: The vehicle users are the main consumers in this system. They hold pseudonyms and related credentials that the OBU makes an agreement with the MA communicating by security channel. After completing the join phase, vehicle users are able to reform their pseudonyms without the MA. The credential related to a pseudonym is minted locally by the vehicle user.
- *Management Authority(MA)*: The MA is responsible for public parameters and membership management. In detail, the MA is supposed to set up the group signature parameters and takes charge of the member tracing and the revocation.
- *RSU*: The RSU is responsible for vehicle road service and identity verification. The RSU holds real-time road information. When the RSU shares road information with the vehicle users, the RSU verifies their identities.
- *Blockchain*: The blockchain is considered as a distributed bulletin board in the proposed system. Due to the tamper

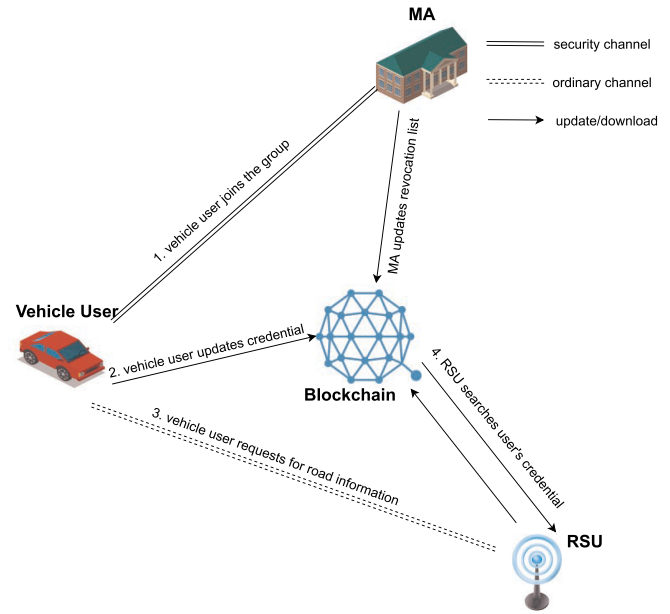


Fig. 2. Overview of proposed system.

resistance, vehicle users and the RSU could have a consistent view of identities. In detail, pseudonyms, credentials, and the revocation list are all uploaded to the blockchain.

B. High-Level Description

As the illustration above, there are four roles in the proposed system. All of them take part in the workflow. Then we will present our system from a high-level description.

Our proposed system is mainly consisting of the pseudonym and the credential. In the pseudonym part, vehicle users could get their keys and generate their own pseudonyms, which are the derivatives of the real identities. In the credential part, vehicle users could form their credentials locally, which are proof of the vehicle user key and attributes. Before these two parts, the vehicle user should run the join parts first. In fact, the local generation leaves much flexibility to the vehicle users.

- *System Setup*: $(pp, msk) \leftarrow \text{Setup}(\mathcal{K})$. This algorithm takes the security parameter $\mathcal{K}$ as input, then it outputs the public parameters pp and the master secret key msk . This process mainly includes the PDN.Setup and Cred.Setup algorithm and it's executed by the management authority for only one time.
- *Vehicle Registration*: $(sk_u) \leftarrow \text{PDN.Join}(pp, id)$. This algorithm takes the public parameters pp , vehicle user's identity information id as input, then it outputs the vehicle user's private key sk_u . This process mainly includes the PDN.Join algorithm and it's executed by both the vehicle user and the management authority. In fact, the management authority actually manages the system.

For the pseudonym part, the group signature could be referred, which could guarantee anonymity.

- *Pdn Generation*: $(pdn) \leftarrow \text{PDN.Gen}(pp, sk_u, m)$. This algorithm takes the public parameters pp , vehicle user's private key sk_u , customized message m as input, then it

outputs the pseudonym pdn . The process mainly includes the PDN.Gen algorithm and it's executed locally by the user.

- *Pdn Verification*: $(b) \leftarrow PDN.Verf(pp, pdn, m)$. This algorithm takes the public parameters pp , vehicle user's pseudonym pdn , customized message m as input, then it outputs a bit $b \in \{0, 1\}$. The process mainly includes the PDN.Verf algorithm and it could be executed by everyone in this system.
- *Revoke*: $(\pi_{revoke}) \leftarrow Rvoke(pp, pdn, m, msk)$. This algorithm takes the public parameters pp , vehicle user's pseudonym pdn , customized message m , management authority's master key msk as input, then it outputs the revocation proof π_{revoke} . The $Rvoke(\cdot)$ process mainly includes the PDN.Open algorithm and the PDN.Revoke algorithm. In detail, it's only executed by the management authority. The PDN.Revoke algorithm is customized and it could realize additional features.

For the credential part, the DAC could be referred. The users form credentials in need, then they could enjoy road services by showing their credentials.

- *Cred Generation*: $(c) \leftarrow Cred.Form(pp, sk_u, attrs)$. This algorithm takes the public parameters pp , vehicle user's private key sk_u , vehicle user's attributes set $attrs$ as input, then it outputs the credential c . The process mainly includes the Cred.Form algorithm and it's executed locally by the vehicle user.
- *Cred Verification*: $(b) \leftarrow Cred.Verf(pp, c, attrs)$. This algorithm takes the public parameters pp , the credential c , vehicle user's attributes set $attrs$ as input, then it outputs a bit $b \in \{0, 1\}$. The process mainly includes the Cred.Verf algorithm and it could be executed by everyone in this system.
- *Cred Show*: $(\pi_s, \pi_v) \leftarrow Cred.Show(pp, sk_u, pdn, c, hint)$. This algorithm takes the public parameters pp , vehicle user's private key sk_u , vehicle user's pseudonym pdn , the hint of valid identity $hint$ as input, then it outputs the proof of show π_s , the proof of valid identity π_v . The process mainly includes the Cred.Show algorithm. In detail, it's executed by the vehicle user.
- *Show Verification*: $(b) \leftarrow ShowVerf(pp, pdn, c, attrs, \pi_s, \pi_v)$. This algorithm takes the public parameters pp , vehicle user's pseudonym pdn , the credential c , vehicle user's attributes set $attrs$, the proof of show π_s , the proof of valid identity π_v as input, then it outputs a bit $b \in \{0, 1\}$. The $ShowVerf(\cdot)$ process mainly includes the Pdn Verification, the Cred Verification and the Cred.ShowVerf algorithm. In detail, it could be executed by everyone in this system.

For blockchain, it provides a place for the other three roles to publish and receive information from each other. As for the MA, he generates user's certificates and admits the user as a group member in the User Registration phase. When some vehicle users become malicious or break some rules, the MA could trace them and even revokes their certificates in the Revoke phase. The RSUs are responsible to verify the validation of the pseudonym and the credential before they share road information. In

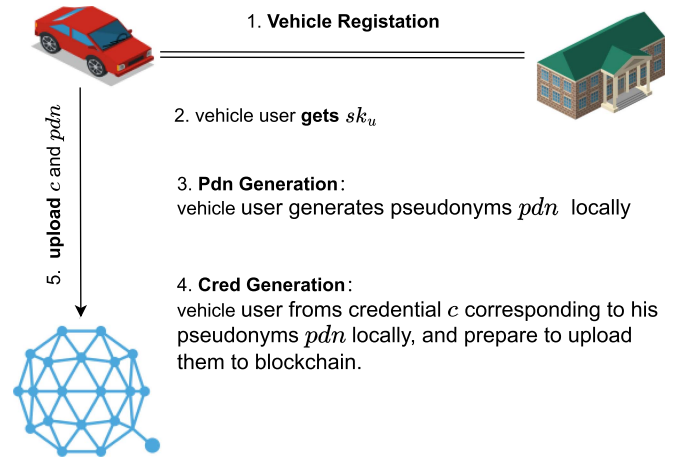


Fig. 3. Registration and credentials forming.

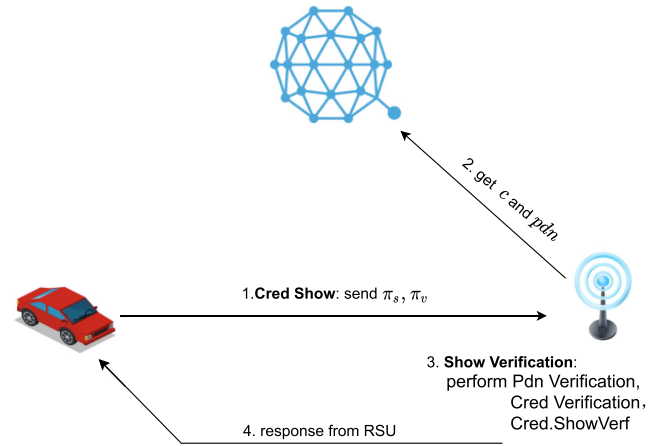


Fig. 4. Request for road information.

addition, it is also necessary to check whether the vehicle user has been revoked.

In practice, the vehicle user is a much active part of the VANET. The vehicle user's activities could be divided into two stages.

In stage one, the vehicle user becomes a valid member from a freshman and gets his credential, as shown in Fig. 3. First of all, the vehicle user is supposed to communicate with the MA using a secure channel for secret keys and certificates in the User Registration phase. The vehicle user will be admitted to the group as a group member if his identity is verified by the management authority. After receiving certificates, he could generate his pseudonym and credential locally. The pseudonym and credential may be uploaded to the blockchain for further usage. The process of pseudonym and credential generation could be performed whenever a vehicle user wants.

In stage two, the vehicle user requests the RSU for road information, as shown in Fig. 4. When a vehicle user is closed to some RSUs, he is supposed to generate proof on the customized message to prove he know some secret values. A path of the Merkle tree is also necessary for membership proof. The RSU gets related credential information from blockchain and verifies information, path, and customized messages according to the

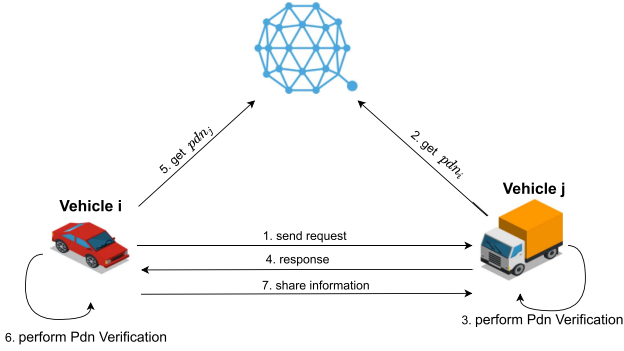


Fig. 5. Information sharing between vehicles.

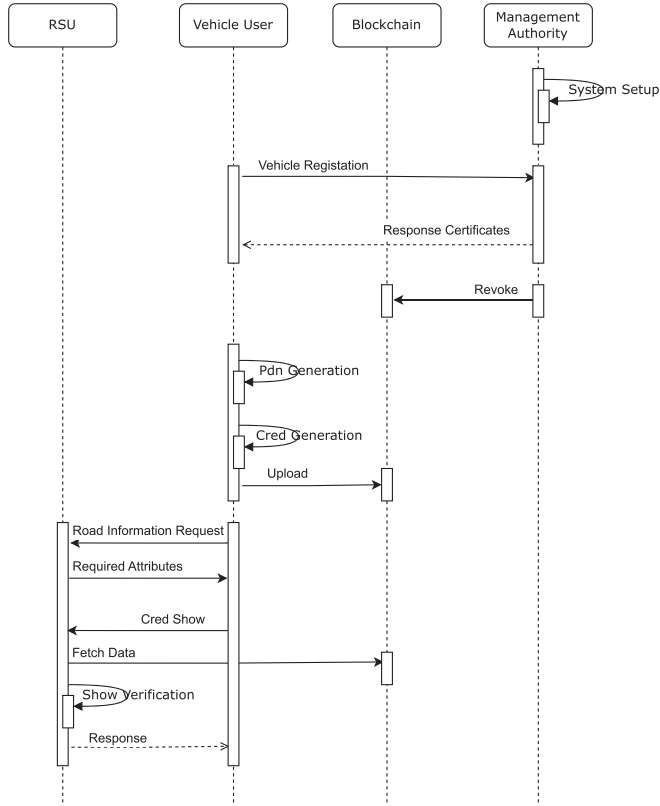


Fig. 6. Sequence of activities in the system.

policy. After all these conditions stand, the RSU responds to the vehicle user about the road information. In addition to the V2I, the V2V communication is also necessary for VANETs and shown in Fig. 5.

To make the process of the proposed system more clear, some activities in the system are shown in Fig. 6. It could be divided into three portions. The first portion is the preparation, where the management authority sets up this system and the vehicle user registers as a group member. Then the second portion is the mintage, where the vehicle user could generate locally his pseudonyms and credentials. Of course, the pseudonyms and credentials will be updated to the blockchain for further usage. The last portion is the consumption, where the vehicle user provides the qualification proof and the RSU verifies these

proofs. Especially, the revocation operated by the management authority is independent of the vehicle user's activities.

From the perspective of the high-level description, the system satisfies the following properties:

- **Anonymity:** Expect the management authority, it is hard for others to distinguish the vehicle user's identity from his pseudonyms and credentials.
- **Unforgeability:** It is hard to forge the pseudonym and credential of a chosen user without knowing his secret key. The valid pseudonym could only be generated by using the valid key. As for the credential, it is formed by a valid vehicle user and is bound with a pseudonym.
- **Traceability:** Given a valid pseudonym, only the management authority could trace the vehicle user's identity hiding in this pseudonym.

C. Threat Models

For the identity management systems for VANET, the coalition of system roles may be an important issue and it could impact the robustness of the system. Even though the management authority is engaged in the identity tracing, it could not be the end of identity forgery if the malicious vehicle users coalesce to forge the vehicle key of an irregular identity. In our system, the management authority is the trusted third party, so we could clearly identify a security and privacy threat model. It is the coalescing malicious vehicle user's security model against the unforgeability for which the coalescing malicious vehicle users attempt to forge a new user key.

We make a more clear description. We say that our system satisfies the unforgeability if for any PPT adversary $\mathcal{A}$, its advantage $Adv_{\mathcal{A}}^{UF}(\mathcal{K})$ in the game $Exp_{\mathcal{A}}^{UF}(\mathcal{K})$ is negligible, where the $\mathcal{K}$ is the security parameter and $Adv_{\mathcal{A}}^{UF}(\mathcal{K}) = |Pr[Exp_{\mathcal{A}}^{UF}(\mathcal{K}) = 1]|$.

The game $Exp_{\mathcal{A}}^{UF}(\mathcal{K})$ is performed between the challenger $\mathcal{C}$ and adversary $\mathcal{A}$. The game is defined as following,

- **Setup:** The challenger $\mathcal{C}$ sets up the system by invoking the System Setup method. Then he gets the public parameter pp and the master secret key msk . The challenger $\mathcal{C}$ keeps msk and sends pp to the adversary $\mathcal{A}$.
- **RegQuery:** The adversary $\mathcal{A}$ makes the User Registration method with the challenger $\mathcal{C}$ for user certificates with user key. For the i -th query, the user key (A_i, e_i) is generated and sent to the adversary $\mathcal{A}$ according to the user key x_i . The total times of queries could not be more than q times.
- **Forge:** The adversary $\mathcal{A}$ forges and outputs the certificate (x^*, A^*, e^*) , where $(x^*, A^*, e^*) \neq (x_i, A_i, e_i)$. The adversary $\mathcal{A}$ wins this game if the forged certificate (x^*, A^*, e^*) passes verification process.

D. Pseudonyms

Referring to the group signature scheme proposed by Giuseppe Ateniese [11] in section III-B, vehicle users' real identities could be covered and pseudonyms could be verified. In the proposed system, an extra revoke phase is developed by using the Merkle tree. The whole phases are presented below:

- **PDN.Setup:** The MA generates $n = pq$ where p, q are two large prime numbers and p, q satisfy with $p = 2p' + 1, q = 2q' + 1$, where p', q' are prime. Select random numbers $a, a_0, g, h \in \mathbb{QR}_n$ and $x \in \mathbb{Z}_{p'q'}^*$, where $\mathbb{QR}_n$ is cyclic subgroup of quadratic residues modulo n . Compute $y = g^x \bmod n$. Finally, the MA publishes the system parameters $params = (n, a, a_0, g, h, y)$ and holds corresponding secret key $secretkey = (p', q', x)$ only known by the MA.
- **PDN.Join:** In this phase, the MA accepts the vehicle user as a group member. Vehicle user's secret key is generated mainly by the user and the user's certificate is generated mainly by the MA. As a result, for i -th vehicle user, his secret key $sk = x_i$ and certificate $cert = (A_i, e_i)$, where $A_i = (a^{x_i} a_0)^{\frac{1}{e_i}} \bmod n$ and prime value e_i is the part of certificate.
- **PDN.Gen:** The vehicle user generates his pseudonym in this phase. In detail, the pseudonym, which is actually consisting of values and proof of knowledge, could be verified by others using system parameters $params$. For i -th vehicle user, his pseudonym $pdn_i = (T_1, T_2, T_3, \pi_p)$, where

$$\begin{aligned}
 T_1 &= A_i y^w, \\
 T_2 &= g^w, \\
 T_3 &= g^{e_i} h^w \\
 \pi_p &= PoK\{(w, e_i, A_i, x_i) : \\
 T_1 &= A_i y^w \\
 T_2 &= g^w \\
 T_3 &= g^{e_i} h^w \\
 A_i &= (a^{x_i} a_0)^{\frac{1}{e_i}} \}
 \end{aligned}$$

and w is randomly chosen by the vehicle user. What's more, we could implement this proof of knowledge (PoK) in the pseudonym referring to the Sigma Protocol and the Fiat-Shamir Paradigm as following

$$\begin{aligned}
 R_1 &= \frac{T_1^{r_1}}{(a^{r_2} y^{r_3})} \\
 R_2 &= \frac{T_2^{r_1}}{g^{r_3}} \\
 R_3 &= g^{r_4} \\
 R_4 &= g^{r_1} h^{r_4} \\
 c_p &= hash(g||h||y||a_0||a||T_1||T_2||T_3||R_1||R_2||R_3||R_4) \\
 s_1 &= r_1 - c_p e_i \\
 s_2 &= r_2 - c_p x_i \\
 s_3 &= r_3 - c_p e_i w \\
 s_4 &= r_4 - c_p w
 \end{aligned}$$

where random numbers $r_1, r_2, r_3, r_4 \in \mathbb{Z}_{p'q'}^*$ and $\pi_p = (c_p, s_1, s_2, s_3, s_4)$.

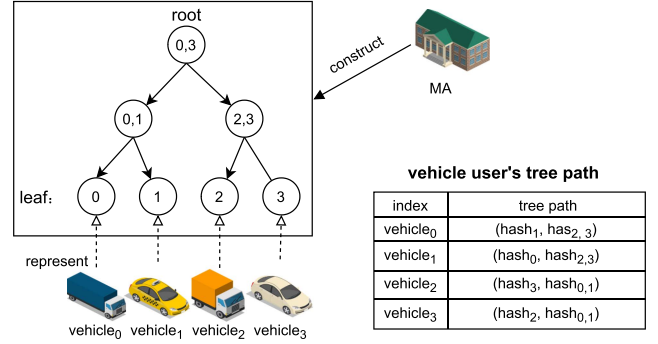


Fig. 7. Merkle tree of membership.

- **PDN.Verf:** The pseudonym could be verified by others. Comparing to the original signature, the message corresponding to the signature may be unnecessary. Since the pseudonym is mainly a proof of knowledge about the certificates, the validation could be processed referring to the $\pi_p = (c_p, s_1, s_2, s_3, s_4)$ as following

$$\begin{aligned}
 R'_1 &= \frac{a_0^{c_p} T_1^{s_1}}{a^{s_2} y^{s_3}} \\
 R'_2 &= \frac{T_2^{s_1}}{g^{s_3}} \\
 R'_3 &= T_2^{c_p} g^{s_4} \\
 R'_4 &= T_3^{c_p} g^{s_1} h^{s_4}
 \end{aligned}$$

and then the proof is valid if the condition $c_p = hash(g||h||y||a_0||a||T_1||T_2||T_3||R'_1||R'_2||R'_3||R'_4)$ holds.

- **PDN.Open:** The MA could trace some vehicle users by opening his signature. For i -th vehicle user, the MA checks the validity of his signature. Then A_i could be recovered by $A_i = \frac{T_1}{T_2^{e_i}}$, where the value x is the secret key of the MA. When publishing this certificate A_i , the MA is supposed to prove that $\log_g y = \log_{T_2} \frac{T_1}{A_i} \bmod n$.
- **PDN.Revoke:** A Merkle tree is constructed for the revocation, as shown in Fig. 7. The leaf node of the Merkle tree is $leaf = Hash(g^{e_i} h^w || Version)$, where e_i is vehicle user's certificate and $Version$ is the version of revocation list. When i -th vehicle user is accepted in the group, his corresponding leaf node is $leaf_i = Hash(g^{e_i} h^w || Version)$. When the MA revokes i -th vehicle user, his corresponding leaf node will be removed from the Merkle tree, as shown in Fig. 8. This figure shows membership proved by using the Merkle tree path.

E. Credentials

The full credentials construction, which we are inspired by Garman et al. [12], is provided below:

- **Cred.Setup:** The MA generates generators $g_0, g_1, \dots, g_t$ of cyclic group $\mathbb{G}$ whose order is n in $params$ of Pseudonyms section.

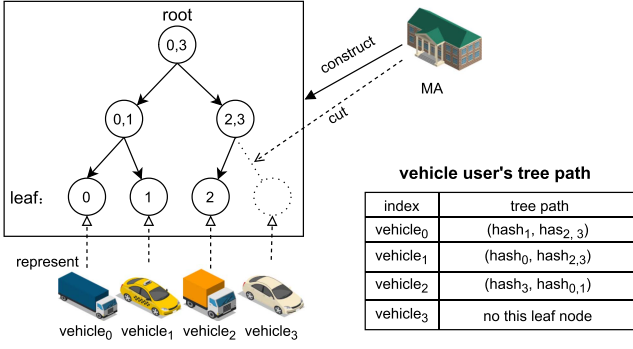


Fig. 8. Revoke member by using merkle tree.

- **Cred.Form:** When a vehicle user generates his pseudonym, he may need a credential for further use. The vehicle user determines his attributes $attrs = (attr_0, attr_1, \dots, attr_{m-1}) \in \mathbb{Z}_n^m$ and auxiliary data aux . Next, the vehicle user selects a random number $w' \in \mathbb{Z}_n$ and computes the credential $c = h^{w'} g^{e_i} \prod_{j=0}^m g_j^{attr_j}$, where e_i is vehicle user's certificate. It outputs (c, π_c) , where π_c is a signature of knowledge on aux that pseudonym and credential contain same value e_i , i.e:

$$\pi_c = SoK[aux] \left\{ (w, e_i, w') : \right.$$

$$T_3 = g^{e_i} h^w$$

$$\wedge c = h^{w'} g^{e_i} \prod_{j=0}^m g_j^{attr_j} \left. \right\}$$

The resulting values $(c, \pi_c, \{g_i^{attr_i}\}, pdn, aux)$ are submitted to blockchain, where pdn is vehicle user's pseudonym. Specially, we could implement this signature of knowledge (SoK) referring to the Sigma Protocol and the Fiat-Shamir Paradigm as following

$$R_1 = g^{r_2} h^{r_1}$$

$$R_2 = h^{r_3} g^{r_2}$$

$$c_c = hash(T_3 || c || R_1 || R_2 || aux)$$

$$s_1 = c_c w + r_1$$

$$s_2 = c_c e_i + r_2$$

$$s_3 = c_c w' + r_3$$

where random numbers $r_1, r_2, r_3 \in \mathbb{Z}_{p'q'}^*$ and $\pi_c = (c_c, s_1, s_2, s_3)$.

- **Cred.Verif:** The verifier accepts this credential iff the proofs π_c and pdn contain the same secret e_i referring to the $\pi_c = (c_c, s_1, s_2, s_3)$ as following

$$R'_1 = g^{s_2} h^{s_1} (T_3)^{-c_c}$$

$$R'_2 = h^{s_3} g^{s_2} \left(\frac{c}{\prod_{j=0}^m g_j^{attr_j}} \right)^{-c_c}$$

and then the proof is valid if the condition $c_c = hash(T_3 || c || R'_1 || R'_2 || aux)$ holds.

- **Cred.Show:** When the vehicle user requires some road information from the RSU, it's necessary for him to show that the required attribute set $\{ra\}$ is in his credentials c and he is not revoked. In fact, the RSUs in different locations may require characteristic attributes. The vehicle user with pdn and c will prove that his corresponding leaf node is in the Merkle tree. The user gets $path$ of his leaf node and makes a signature of knowledge π_s on $\{ra\}$, i.e:

$$\pi_s = SoK \left\{ (w, e_i, \{ra\}) : \right.$$

$$c = h^w g^{e_i} \prod_{i \in \{ra\}} g_i^{ra_i} \prod_{j \in \{attr\} \setminus \{ra\}} g_j^{attr_j} \left. \right\}$$

Then the user sends $(path, \pi_s, g^{e_i} h^w, \{attr\} \setminus \{ra\})$ to the RSU. This phase may take place in communication between the vehicle user and the RSU. We could implement this signature of knowledge referring to the Sigma Protocol and the Fiat-Shamir Paradigm in the same way as following

$$R = h^{r_w} g^{r_e} \prod_{i \in \{ra\}} g_i^{ra_i}$$

$$c_s = hash(c || R)$$

$$s_w = c_s w + r_w$$

$$s_e = c_s e_i + r_e$$

$$s_1 = c_s ra_1 + r_1$$

$$s_2 = c_s ra_2 + r_2$$

$$\dots$$

$$s_m = c_s ra_m + r_3$$

where m is the size of the set $\{ra\}$, random numbers $r_w, r_e, r_1, r_2, \dots, r_m \in \mathbb{Z}_{p'q'}^*$, $ra_i \in \{ra\}$ and $\pi_s = (c_s, s_w, s_e, s_1, s_2, \dots, s_m)$.

- **Cred.ShowVerf:** For the verifier, he finds corresponding (c, π_c, pdn, aux) on blockchain and receives $(path, \pi_s, g^{e_i} h^w, \{attr\} \setminus \{ra\})$ from the vehicle user. If $path$ holds in the Merkle tree and π_s is valid, the verifier will believe that the required attribute set $\{ra\}$ is included in c and the vehicle user is not revoked. In detail, the verification process of π_s is presented as following

$$R' = h^{s_w} g^{s_e} \prod_{i \in \{ra\}} g_i^{s_i} \left(\frac{c}{\prod_{j=0}^m g_j^{attr_j}} \right)^{-c_s}$$

and then the proof is valid if the condition $c_s = hash(c || R')$ holds.

V. SECURITY ANALYSIS

A. Correctness

We will give the correctness analysis of our proposed system as following:

- *Pseudonym*: After the registration process, it is the first step for vehicle user i to generate the pseudonym with the user key (A_i, e_i) . Assume the pseudonym is $pdn_i = (T_1, T_2, T_3, \pi_p)$, where

$$T_1 = A_i y^w \mod n,$$

$$T_2 = g^w \mod n,$$

$$T_3 = g^{e_i} h^w \mod n,$$

and w is random. In detail, for the verifier, he could get the values (R'_1, R'_2, R'_3, R'_4) with the pdn_i and check the $c_p = \text{hash}(g||h||y||a_0||a||T_1||T_2||T_3||R'_1||R'_2||R'_3||R'_4)$ if the following equations stands.

$$\begin{aligned} R'_1 &= a_0^{c_p} T_1^{s_1} / (a^{s_2} y^{s_3}) \\ &= a_0^{c_p} T_1^{r_1 - c_p e_i} / (a^{r_2 - c_p x_i} y^{r_3 - c_p e_i w}) \\ &= a_0^{c_p} T_1^{r_1} T_1^{-c_p e_i} / (a^{r_2} a^{-c_p x_i} y^{r_3} y^{-c_p e_i w}) \\ &= a_0^{c_p} T_1^{r_1} (A_i y^w)^{-c_p e_i} / (a^{r_2} a^{-c_p x_i} y^{r_3} y^{-c_p e_i w}) \\ &= a_0^{c_p} T_1^{r_1} (A_i)^{-c_p e_i} y^{-c_p e_i w} / (a^{r_2} a^{-c_p x_i} y^{r_3} y^{-c_p e_i w}) \\ &= a_0^{c_p} T_1^{r_1} ((a^{x_i} a_0)^{\frac{1}{e_i}})^{-c_p e_i} / (a^{r_2} a^{-c_p x_i} y^{r_3}) \\ &= a_0^{c_p} T_1^{r_1} (a^{-c_p x_i} a_0^{-c_p}) / (a^{r_2} a^{-c_p x_i} y^{r_3}) \\ &= T_1^{r_1} / (a^{r_2} y^{r_3}) \\ &= R_1 \\ R'_2 &= T_2^{s_1} / g^{s_3} \\ &= T_2^{r_1 - c_p e_i} / g^{r_3 - c_p e_i w} \\ &= T_2^{r_1} T_2^{-c_p e_i} / g^{r_3} g^{-c_p e_i w} \\ &= T_2^{r_1} (g^w)^{-c_p e_i} / g^{r_3} g^{-c_p e_i w} \\ &= T_2^{r_1} g^{-c_p e_i w} / g^{r_3} g^{-c_p e_i w} \\ &= T_2^{r_1} / g^{r_3} \\ &= R_2 \\ R'_3 &= T_2^{c_p} g^{s_4} \\ &= T_2^{c_p} g^{r_4 - c_p w} \\ &= (g^w)^{c_p} g^{r_4 - c_p w} \\ &= g^{w c_p} g^{r_4 - c_p w} \\ &= g^{r_4} \\ &= R_3 \\ R'_4 &= T_3^{c_p} g^{s_1} h^{s_4} \\ &= T_3^{c_p} g^{r_1 - c_p e_i} h^{r_4 - c_p w} \end{aligned}$$

$$\begin{aligned} &= (g^{e_i} h^w)^{c_p} g^{r_1 - c_p e_i} h^{r_4 - c_p w} \\ &= g^{e_i c_p} h^{c_p w} g^{r_1 - c_p e_i} h^{r_4 - c_p w} \\ &= g^{r_1} h^{r_4} \\ &= R_4 \end{aligned}$$

- *Credential*: The credential should be formed with the pseudonym. Assume the credential is $c = h^{w'} g^{e_i} \prod_{j=0}^m g_j^{attr_j}$, where e_i is the vehicle user key and w' is random. The credential c includes the vehicle user key and vehicle user's attributes, so the correctness process of credential verification is to prove that the credential and pseudonym both include the same user key. The proof π_c guarantees this binding relationship. In detail, for the verifier, he could get the values (R'_1, R'_2) with the π_c and check the $c_c = \text{hash}(T_3||c||R'_1||R'_2||aux)$ if the following equations stands.

$$\begin{aligned} R'_1 &= g^{s_2} h^{s_1} (T_3)^{-c_c} \\ &= g^{c_c e_i + r_2} h^{c_c w + r_1} (g^{e_i} h^w)^{-c_c} \\ &= g^{c_c e_i + r_2 - c_c e_i} h^{c_c w + r_1 + c_c w} \\ &= g^{r_2} h^{r_1} \\ &= R_1 \\ R'_2 &= h^{s_3} g^{s_2} \left(\frac{c}{\prod_{j=0}^m g_j^{attr_j}} \right)^{-c_c} \\ &= h^{c_c w' + r_3} g^{c_c e_i + r_2} \left(\frac{c}{\prod_{j=0}^m g_j^{attr_j}} \right)^{-c_c} \\ &= h^{c_c w' + r_3} g^{c_c e_i + r_2} (h^{w'} g^{e_i})^{-c_c} \\ &= h^{r_3} g^{r_2} \\ &= R_2 \end{aligned}$$

- *Show*: The credential c includes the attributes, which may be required by the RSU. The correctness process of show verification is to prove that some attribute set in the credential meets the requirement. In order to present the required attribute set hiding in the credential c , the proof π_s is necessary. The proof π_s guarantees that the required attributes exist in the credential c and they are qualified with the requirement of the RSU. In the construction of the proof π_s , the Sigma Protocol and the Fiat-Shamir Paradigm could be referred to. In detail, for the verifier, he could get the values R' with the π_s and check the $c_c = \text{hash}(c||R')$ if the following equations stands.

$$\begin{aligned} R' &= h^{s_w} g^{s_e} \prod_{i \in \{ra\}} g_i^{s_i} \left(\frac{c}{\prod_{j \in \{attr\} \setminus \{ra\}} g_j^{attr_j}} \right)^{-c_s} \\ &= h^{c_s w + r_w} g^{c_s e_i + r_e} \prod_{i \in \{ra\}} g_i^{c_s r_{a_i} + r_i} \left(h^w g^{e_i} \prod_{i \in \{ra\}} g_i^{r_{a_i}} \right)^{-c_s} \end{aligned}$$

$$\begin{aligned}
&= h^{r_w} g^{r_e} \prod_{i \in \{ra\}} g_i^{r_i} \\
&= R
\end{aligned}$$

- **Revocation:** As for the management authority, he is supposed to revoke malicious users and update the revocation tree. The correctness process of revocation verification is the checking of the integrity in the Merkle tree. As shown in Figs. 7 and 8, only the unpruned leaves have the hash path. Since the management authority could trace the vehicle user i and prune the corresponding leaf $leaf_i = Hash(g^{e_i} h^w || Version)$, the revoked vehicle user i will not be valid by verifying the hash path. This process could be complete when the new revoked tree root is published in a block.

B. Unforgeability

Theorem 1: If the unforgeability in the work [11] holds, then the certificate in our scheme is unforgeable.

Proof.

Assume that there is a P.P.T adversary $\mathcal{A}$ that could win the game $Exp_A^{UF}(\mathcal{K})$ with a non-negligible possibility ϵ , then we could break the unforgeability of the work [11] with a non-negligible possibility ϵ' by taking advantage of the adversary $\mathcal{A}$.

For convenience, we denote the init and join process in the work [11] by $\mathcal{O}_{init}$, $\mathcal{O}_{join}$ respectively, where $pp \leftarrow \mathcal{O}_{init}(\mathcal{K})$, $(A, e) \leftarrow \mathcal{O}_{join}(pk_u)$ and the pk_u is the user's part public key.

- **Setup:** The challenger $\mathcal{C}$ get the public parameters pp by querying the oracle $\mathcal{O}_{init}$. Then $\mathcal{C}$ sends pp to the adversary $\mathcal{A}$.
- **RegQuery:** This query could not be more than q times. For the i -th query, the adversary $\mathcal{A}$ sends pk_i to the challenger $\mathcal{C}$ as the user's part public key. Then the challenger $\mathcal{C}$ query the $\mathcal{O}_{join}$ with pk_i and obtains the output (A_i, e_i) as the certificate. The challenger $\mathcal{C}$ sends (A_i, e_i) to the adversary $\mathcal{A}$ as response.
- **Forge:** The adversary $\mathcal{A}$ forges and outputs the certificate (x^*, A^*, e^*) , where $(x^*, A^*, e^*) \neq (x_i, A_i, e_i)$. The challenger $\mathcal{C}$ outputs (x^*, A^*, e^*) if the forged certificate (x^*, A^*, e^*) passes verification process. Otherwise, the challenger terminates and outputs failure.

We consider the possibility to break the unforgeability in the work [11]. Actually there are events considered for challenger $\mathcal{C}$ as following:

- Pr_1 . As there is no termination in the RegQuery phase, the probability of event that it doesn't abort in the RegQuery phase, which is $Pr_1 \geq 1$.
 - Pr_2 . The probability of event that adversary $\mathcal{A}$ wins the game $Exp_A^{UF}(\mathcal{K})$, which is $Pr_2 \geq \epsilon$.
 - Pr_3 . The probability of event that adversary $\mathcal{A}$ choose useful attack in the forgery phase, which is $Pr_3 \geq 1$
- then we believe challenger $\mathcal{C}$ break the work if the above events happen simultaneously:

$$\epsilon' = Pr_1 Pr_2 Pr_3$$

$$\geq \epsilon$$

□

C. Signature of Knowledge

Theorem 2: If the EUF-CMA holds in the Schnorr signature scheme, then signature of knowledge π_c satisfies *correctness*, *special soundness* and *honest verifier zero-knowledge*.

Proof: Correctness. The proof process about the correctness has been presented in part Section V-A.

Special Soundness. About the process of proof, if the prover could convince the verifier, there could be an extractor that extracts the knowledge about the proof. The soundness implicates the existing knowledge, which protects the verifier against the cheating of the prover.

Taking advantage of the rewinding technology, the extractor could get two transcripts about the proof with the same random value (r_1, r_2, r_3) . Assume these transcripts are $(r_1, r_2, r_3, c_c, s_1, s_2, s_3)$ and $((r_1, r_2, r_3, c'_c, s'_1, s'_2, s'_3))$ respectively. Then the knowledge could be extracted as following:

$$w = \frac{s'_1 - s_1}{c'_c - c_c}$$

$$e = \frac{s'_2 - s_2}{c'_c - c_c}$$

$$w' = \frac{s'_3 - s_3}{c'_c - c_c}$$

Honest Verifier Zero-Knowledge. To present the process, we assume there is a simulator $\mathcal{S}$, which could generate a simulated transcript by giving the random c_c . For the verifier, he could not distinguish the real transcript and the simulated transcript.

The generation process of the simulator $\mathcal{S}$ is presented as following with the given random c_c :

- 1) The simulator $\mathcal{S}$ generates random numbers (s_1^r, s_2^r, s_3^r) .
- 2) The simulator $\mathcal{S}$ computes values as following:

$$R_1^r = g^{s_1^r} h^{s_2^r} (T_3)^{-c_c}$$

$$R_2^r = h^{s_3^r} g^{s_2^r} \left(\frac{c}{\prod_{j=0}^m g_j^{attr_j}} \right)^{-c_c}$$

- 3) The simulator $\mathcal{S}$ outputs the simulated transcript $(R_1^r, R_2^r, c_c, s_1^r, s_2^r, s_3^r)$

□

D. System Goals Analysis

In this paper, we could find that (1) the vehicle user could generate pseudonyms and credentials locally, which makes it possible for vehicles to prepare the pseudonyms and corresponding credentials in reserve; (2) The revocation with the Merkle tree might be affordable to management authority while there is no update operation leaving to vehicle users; (3) Most existing blockchain systems could be qualified and there are no prerequisites for system deployment, since the roles in system are merely uploading and fetching the required data.

In addition, the proposed system should satisfy following properties, which include those mentioned in Section IV-B.

- **Anonymity:** The vehicle users' identities, which is included in the pseudonym and the credentials, will not be revealed when the pseudonym and credentials are uploaded to the public blockchain. What's more, others could learn nothing

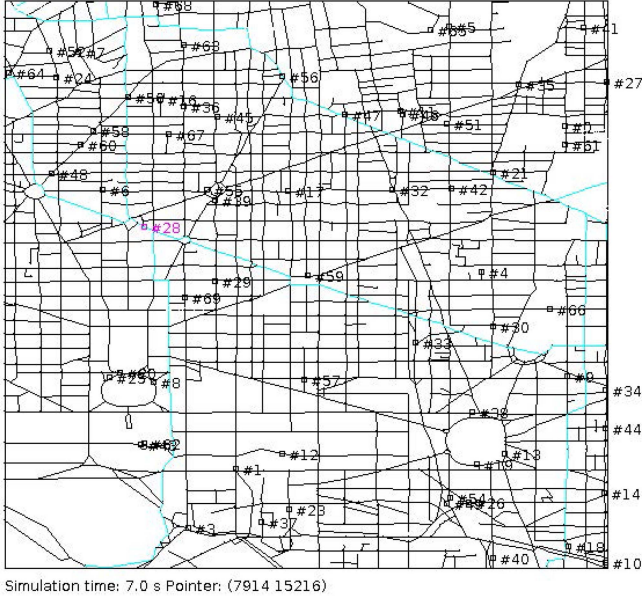


Fig. 9. Simulation of vehicles by ns2.

more than the selected attributes in anonymous credentials on the blockchain. The group signature verification process doesn't reveal the identity of the vehicle user. In the show phase, g^{e_i} may be exposed to the RSU, however, e_i is still inaccessible. As anonymity stands, the identity privacy of vehicle users could be realized.

- **Traceability:** The valid pseudonym is $p_{dn} = (T_1, T_2, T_3, \pi_p)$, where $T_1 = A_i y^w \bmod n$ and $T_2 = g^w \bmod n$. For the management authority, who holds the secret x , he could get $A_i = \frac{T_1}{T_2^x}$ easily, where A_i is the part identity of vehicle user i . As for location privacy, the V2V only involves pseudonyms that hold anonymity according to group signature. It may provide approaches to preserve location privacy since the vehicle users could form pseudonyms and credentials as required.
- **Unforgeability:** The valid certificate could be only issued by the management authority. The certificate $cert = (A_i, e_i)$ is necessary for the generation of pseudonyms and credentials, which makes it hard to forge pseudonyms and credentials. What's more, according to the Theorem 1, our system holds unforgeability.

VI. PERFORMANCE EVALUATION

In this section, we will evaluate the performance of the proposed system. Since our system has few requirements for blockchain, we will mainly concentrate on the functions and algorithms in the proposed system.

We evaluate these computations corresponding to the denotation. The performance is made on a personal computer (with Intel(R) Core(TM) i7-7700 CPU @3.60GHz(3600 MHz), Windows 10, 8 GB memory) by using the MIRACL library. In detail, the running time is an average value of 1000 times execution in 128-bit security level. We simulated the VANETs environment by using ns2 tools, as shown in Fig. 9.

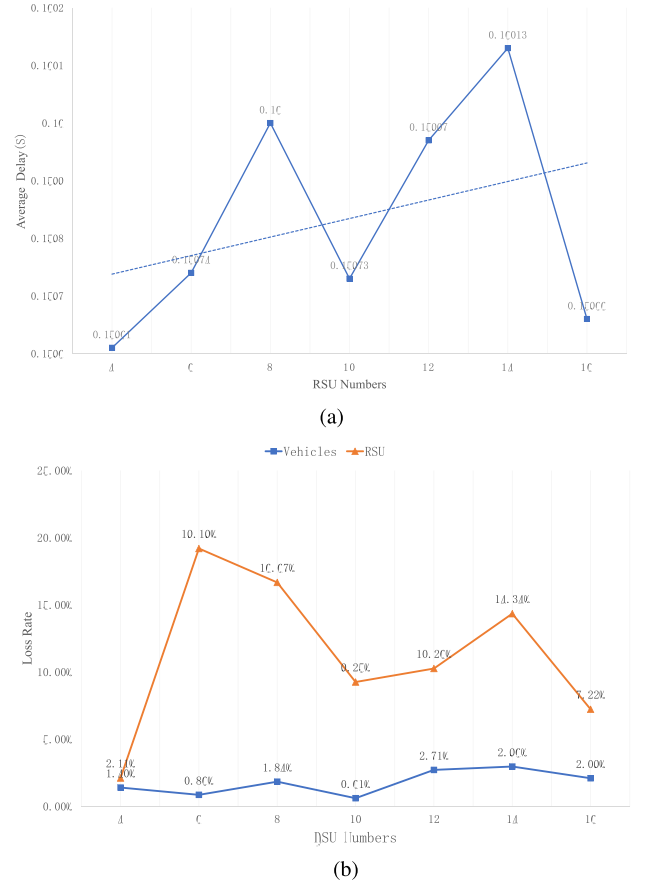


Fig. 10. Experiments results with 70 vehicles and 30 km/h. (a) The average delay with the RSU number increasing. (b) The loss rate with the RSU number increasing.

We conduct the test of RSU nodes. The speed is set at 30 km/h and the number of vehicle nodes is 70. Then we set the number of RSU nodes at 2, 4, 6, 8, 10, 12, 14 and 16 respectively to test the influence of the RSU on the delay and the loss rate.

As shown in Fig. 10(a), the average delay of vehicle nodes keeps on changing when RSU nodes are increased. The fluctuation range and specific values of the test results show that communication congestion is not the main factor affecting the delay when RSU nodes are increased. Therefore, the processing of the scheme protocol is the main factor. The Fig. 10(b) shows an obvious change in the packet loss rate of the system. The packet loss rate of RSU nodes holds a decreasing trend, which is consistent with the expectation. Multiple RSUs release the computing pressure of RSU groups to respond to vehicle node requests.

Finally we perform the next experiment. We set the vehicle speed at 30 km/h, the number of RSU nodes at 16, and the vehicle nodes at 70, 80, 100, 120, 150, 180, 200 and 250 respectively for testing.

The Fig. 11(a) shows the rising trend of the average request delay, and it can be seen from this trend that the influence of communication queue on vehicle request delay is gradually increasing. The packet loss rate of system nodes reflects a bigger problem, shown in the Fig. 11(b) The packet loss rate shows a

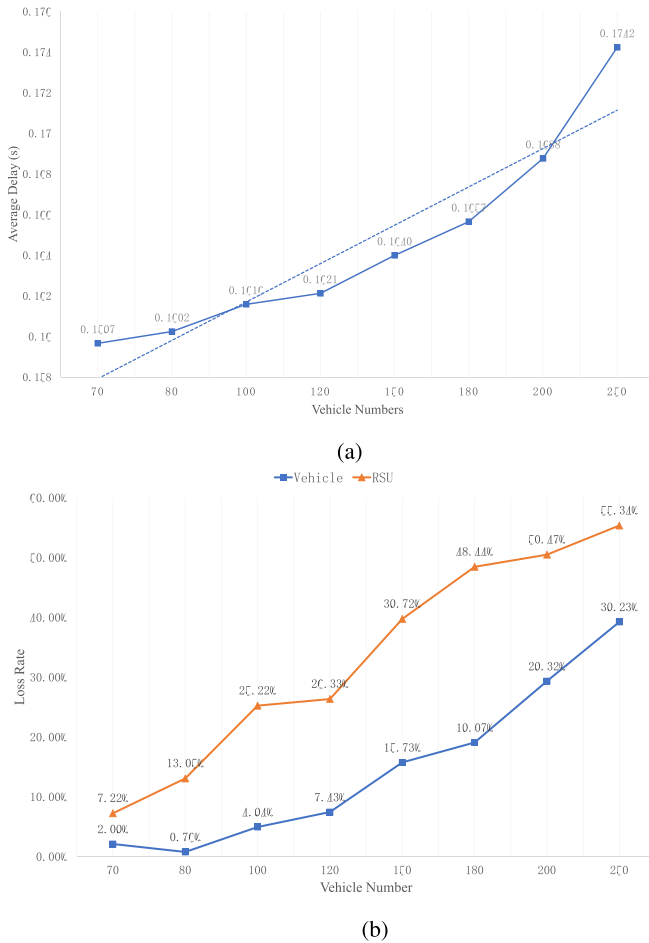


Fig. 11. Experiments results with 16 RSUs and 30 km/h. (a) The average delay with the vehicle number increasing. (b) The loss rate with the vehicle number increasing.

very obvious upward trend, and when the vehicle node reaches 250, even if the RSU node almost covers the entire simulation map, the vehicle packet loss rate is close to 40%, and the RSU packet loss rate is 55%.

There is a special phenomenon in the above experimental tests. When the number of vehicle nodes is above 120, each evaluation index increases significantly. In view of this phenomenon, the communication intensity may exceed the bottleneck of the system when the vehicle node is at 120. What's more, when the RSU node is at 16 and the vehicle speed is 30 km/h on the simulation map, the simulation system in this paper reaches a relatively full state when the vehicles node is at 120. For the functional consideration, the system achieves the identity privacy protection during the whole process of anonymity and revocation.

VII. CONCLUSION

Identity privacy and location privacy are the core privacy-preserving requirements in VANET. As a significant identity management technology, decentralized anonymous credentials could be a practical approach to preserve the privacy of vehicle users. Blockchain, as a valuable technology, could provide anonymous credentials systems with infrastructure,

which makes the process of credential generation and verification trusted. Meanwhile, existing anonymous credentials systems have limitations in architecture and functionality. Anonymity may lead to inconvenience in traceability and revocation, while these two properties are necessary for the practical application of VANET. As for location privacy, the leakage could be relieved by reasonable usage of pseudonyms and credentials. What's more, efficient revocation is a much practical requirement in this environment, as misbehavior is supposed to be stopped to protect the property.

Our proposed system remains the management authority as a trusted third party to simplify credential issuance, traceability, and revocation, while blockchain is also employed as a consistent bulletin board for public verification. In order to show our advantage, we make kinds of experiments with the car numbers and the RSU numbers to obtain the performance of the system. As a result, our system performs well in the show and revocation phase and holds flexible privacy functions.

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